

Student Answer Sheet

Subject: Artificial Intelligence

Definition:

Artificial Intelligence is the branch of computer science that enables machines to perform tasks that would normally require human intelligence. This includes learning from experience, understanding language, recognising patterns, and making decisions. AI systems use algorithms and data to simulate intelligent behaviour.

Body:

AI includes Machine Learning where systems learn from data, Deep Learning which uses layered neural networks for complex pattern recognition, and NLP which processes human language. Computer Vision allows machines to interpret images. Applications include medical diagnosis in healthcare, fraud detection in finance, and autonomous vehicles in transportation. Techniques include supervised, unsupervised, and reinforcement learning.

Conclusion:

AI is transforming industries and offers significant benefits in automation and efficiency. However, challenges such as bias, privacy violations, and job displacement must be managed. Ethical and responsible AI development with transparency and fairness is crucial for positive societal outcomes.